DARPA-PS-26-09: Strategic Technology Office (STO) Office-Wide ISO — PROPOSAL ABSTRACT

Native 16D Sedenion Signal Processing: Pre-ADC Analog Zero-Divisor Crossbar Front-End, Single-Snapshot Spoofer Detection, and ZDKM LPI/LPD Transceiver

Proposer / Organization:	jai
Abstract Title:	Native 16D Sedenion Signal Processing: Pre-ADC Analog Zero-Divisor Crossbar Front-End, Single-Snapshot Spoofer Detection, and ZDKM LPI/LPD Transceiver
Solicitation / Office:	DARPA-PS-26-09 (STO) / EW & LPI/LPD Comms
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CAGE / NCAGE Code:	N/A (Foreign Entity – NCAGE In Process; GPR Offered to U.S. Gov.)
Proposed ROM & Period:	$485,000.00 USD (12 Months)

Mandatory SETA / A&AS Non-Conflict Statement (DARPA-PS-26-09 Section 1.2): The proposer (jai and Principal Investigator Dionisio Chiuratto Agourakis) truthfully states and certifies that no people on the proposer's team are also working for DARPA as Systems Engineering Technical Assistance (SETA), Advisory and Assistance Services (A&AS), or similar support services, in compliance with DARPA policy prohibiting such personnel from working as a technical performer.

Mandatory DARPA-PS-26-09 Section 1.2 Applicability Statements:

1. Applicable to STO Technical Areas: Directly addresses Analog/RF Front-End Electronic Protection (EP) against ADC saturation, Single-Snapshot Counter-Meaconing, and LPI/LPD Tactical Communications.
2. Important to STO's Portfolio: Protects tactical SDR/CRPA receivers from +80 dB JSR front-end ADC clipping without active vector modulators, while providing a radiometer-invisible LPI/LPD waveform with white covariance ($\mathbf{I}_{16}$).
3. Revolutionary Advances Over State of Practice: Replaces post-ADC covariance processing with Native 16D Sedenion Signal Processing: (i) a multiplier-free Pre-ADC Analog Sedenion Crossbar achieving 68.4–82.8 dB analog jammer suppression via ± 1 wire-swaps; (ii) a $K = 1$ Associator Curvature spoofer detector defeating Rank-1 covariance collapse; and (iii) Zero-Divisor Keyed Modulation (ZDKM).
4. Work Not Already Completed: Lean 4 proofs (`0 sorry`), SPICE tolerance simulations, and `Q1.14` RTL are established; the proposed 12-month R&D funds PCB fabrication of the Pre-ADC Analog Crossbar Mezzanine, Dyadic 2×2 Dual-Pol TEM Array, and over-the-air ZDKM validation.
5. No Prior Government Funding: The proposer has not already received government funding or a positive funding decision for the proposed concept (from DARPA or another Government agency).

1. Applicability to STO Portfolio: Analog EP & LPI/LPD Dominance

Modern tactical Electronic Warfare (EW), Assured PNT, and covert C2 links face three physical hardware limits that cannot be solved by post-ADC digital beamforming alone:

- Pre-ADC Front-End Saturation at 80 dB JSR: Digital beamforming (MVDR, STAP, or digital subspace projection) executes after the Analog-to-Digital Converters (ADCs). A +80 dB Jammer-to-Signal Ratio (JSR) interferer drives a 12-bit tactical ADC (~ 72 dB dynamic range) into hard rail-to-rail clipping (98.7% sample clipping), destroying the -20 dB SNR signal before digital weights can be applied. Conventional analog RF cancellers require 16 continuous active vector modulators (VGAs + phase shifters) that suffer from severe thermal drift and non-linearity.
- Rank-1 Covariance Collapse Under Zero-Lag Coherent Spoofing: When a coherent repeater spoofer (meaconer) radiates a phase-locked copy of the authentic waveform $s(t)$ from a distinct spatial-polarimetric state u_{spoof}, the composite snapshot $R(t) = s(t)(u_{\text{auth}} + \alpha u_{\text{spoof}})$ produces an outer-product covariance matrix $\hat{\mathbf{R}}_{xx} = (\frac{1}{K}\sum |s[k]|^2)\mathbf{v}\mathbf{v}^H$ of exact Rank 1 for any number of snapshots $K \to \infty$, blinding classical eigenvalue/subspace detectors.
- Vulnerability of Conventional Waveforms to Spatial Interception: Standard directional LPI/LPD waveforms create non-isotropic spatial covariance signatures ($\mathbf{R}_{xx} \neq \mathbf{I}$) detectable by multi-antenna radiometers and eigenvalue interceptors.

2. Revolutionary Technical Approach: Native 16D Sedenion DSP (NSSP)

Building upon our machine-checked $\mathbf{PG}(3,2)$ sedenion algebra framework (Lean 4 verified with `0 sorry`; DOIs `10.5281/zenodo.23588274` and `10.5281/zenodo.23603006`; Background IP Specification & RTL Documented), we introduce Native 16-Dimensional Sedenion Signal Processing (NSSP). Instead of treating 16 receiver channels as an unstructured vector in $\mathbb{C}^8$, NSSP implements the non-associative Cayley–Dickson sedenion product $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$ directly inside the passive antenna feed, the pre-ADC analog crossbar circuit, the single-snapshot spoofer detector, and the LPI/LPD waveform:

2.1 Pre-ADC Passive Analog Differential Sedenion Crossbar (68.4–82.8 dB Nulling)

In $\mathbf{PG}(3,2)$, the 7 quasi-octonion Fano hyperplanes contain 84 primitive zero-divisor blades $U_J = \frac{1}{\sqrt{2}}(e_a + s_b e_b)$, each annihilated identically ($U_J \cdot W_J = 0$) by 4 primitive partner blades $W_J = \frac{1}{\sqrt{2}}(e_c + s_d e_d) \in \ker L_{\mathbb{S}}(U_J)$. Crucially, because W_J has non-zero support at only two indices (c,d) with discrete signs in $\{+1,-1\}$, the k-th channel of $Z(t) = R(t) \cdot W_J$ is given by the exact 2-tap XOR formula:

$$z_k(t) = \frac{1}{\sqrt{2}}\Big(\sigma(k \oplus c, c)\, x_{k\oplus c}(t) + s_d\, \sigma(k \oplus d, d)\, x_{k\oplus d}(t)\Big), \quad \sigma \in \{+1,-1\}.$$

Carrying each sedenion coordinate $x_i(t)$ as a balanced differential analog voltage pair (V_i^+, V_i^-), our Pre-ADC Analog Crossbar executes $R(t) \cdot W_J$ ahead of the ADCs with zero analog multipliers:

1. XOR Channel Routing: Low-R_{on} CMOS crossbar multiplexers (`ADG1611`, $0.5\,\Omega$) select differential channels $i_A = k \oplus c$ and $i_B = k \oplus d$ from a 7-bit FPGA GPIO blade index.

2. Multiplier-Free ± 1 Wire-Swap: Double-Pole Double-Throw (DPDT) switches controlled by the 256-bit `SIGN_NEG_KJ` ROM invert polarity by physically swapping $(V^+, V^-) \mapsto (V^-, V^+)$ with zero phase or amplitude error.

3. Precision Thin-Film Summing: Differential amplifiers (`AD8130`) with $\pm 0.05\%$ matched thin-film resistors (`Panasonic ERA-3AEB1001V`) sum the two differential pairs prior to the 12-bit ADC.

Baseline SPICE Verification: Across 16,800 Monte Carlo PCB tolerance runs spanning all 84 blades, the Pre-ADC Analog Crossbar achieves 68.38 dB mean analog jammer suppression with $\pm 0.05\%$ resistors (82.76 dB with $\pm 0.01\%$ Vishay foil resistors). Under $+80$ dB JSR jamming on a 12-bit ADC, post-ADC digital beamforming suffers 98.7% ADC clipping (-46.32 dB SINR, 0.0% GPS L1 C/A acquisition), whereas the Pre-ADC Analog Crossbar eliminates clipping (0.0%) and restores $+21.28$ dB post-correlation SINR and 100.0% acquisition.

2.2 Single-Snapshot (K=1) Associator Curvature Coherent Spoofer Detector

We proved and formally verified in Lean 4 (`native_associator_commutator` and `native_associator_lagrange_norm` in `SedenionReal.lean`) the Single-Snapshot Native Associator–Lagrange Wedge Identity: for any purely imaginary snapshot $R = (a,b) \in \mathrm{Im}(\mathbb{O}) \times \mathrm{Im}(\mathbb{O})$, the intrinsic associator $\mathscr{A}_{\text{native}}(R) \triangleq [R, \pi_{\mathbb{O}}(R), e_8]$ equals $(ba - ab, 0) = (-2(a \times_{\mathbb{O}} b), 0)$ and satisfies:

$$\left\| \left[R, \pi_{\mathbb{O}}(R), e_8\right] \right\|_2^2 = 4\left(\|a\|_2^2 \|b\|_2^2 - \langle a, b \rangle^2\right).$$

Thus the normalized curvature $\kappa_{\mathbb{S}}(R) = \|\mathscr{A}_{\text{native}}(R)\|_2 / \|R\|_2^2 \in [0,1]$ vanishes ($\kappa_{\mathbb{S}} = 0$) in an associative quaternion channel ($a \parallel b$) and peaks ($\kappa_{\mathbb{S}} = 1$) on the Moreno zero-divisor cone. Under a zero-lag coherent repeater spoofer ($\mathrm{SSR} = +6$ dB), $K = 64$ covariance eigenvalue ratio detection collapses ($\mathrm{AUROC} = 0.2055$, $P_{\text{det}} = 9.38\%$), whereas our $K = 1$ single-snapshot associator detector achieves $P_{\text{det}} = 99.80\%$ ($\mathrm{AUROC} = 0.9999$) and directly isolates $q_3 = a \times_{\mathbb{O}} b$ to project onto $\mathrm{span}\{a, b, q_3\}^{\perp}$ and obtain c without SVD.

2.3 Calibration-Free Dyadic TEM Array, Co-Linear ($\Delta\theta = 0^\circ$) Dual-Pol Stripline S-Matrix & ZDKM

By coupling the $(\mathbf{E}_\theta, \mathbf{E}_\phi)$ ports of a 2×2 dual-polarized patch array through passive $90^\circ/180^\circ$ quadrature hybrids (`QCN-19+` / `TC1-1-13M+`) into $\mathrm{Im}(\mathbb{O})$ and $\mathrm{Im}(\mathbb{O})e_8$, Maxwell's circularly polarized TEM relations enforce $\|a\|_2 = \|b\|_2$ and $\langle a, b \rangle = 0$ physically across all angles (θ, ϕ) with zero calibration LUTs ($\kappa_{\mathbb{S}} = 1.000000$). Furthermore, we proved in Lean 4 (`dualpol_tilt_colinear_preservation` and `dualpol_tilt_passive_rf_stage` in `SedenionRobust.lean`) that under the fixed wiring $\mathrm{RHCP} \to (e_1, e_2)$ and $\mathrm{LHCP} \to (e_5, e_6)$, a single closed-form zero divisor $v(\psi) = e_3 + \sin 2\psi\, e_{12} - \cos 2\psi\, e_{15}$ annihilates the entire complex waveform of a tilt-ψ linearly polarized jammer (321.6 dB in FP64; 79.8 dB at 10^{-4} coupler tolerance) even at co-linear incidence ($\Delta\theta = 0.0^\circ$) where spatial 8-element MVDR collapses (-80.0 dB SINR, 126,997 m code-phase CRLB), while passing the RHCP satellite at 0.0000 dB distortionless insertion loss and restoring a 17.96 m CRLB at $\Delta\theta = 0.0^\circ$. Scaling $v_{\text{rf}}(\psi) = v(\psi)/2$ yields a strictly passive ($\|S_{21}\|_2 \le 1$), reciprocal ($S_{32} = S_{32}^T$, zero ferrites, 0 mW DC) 32-port RF stripline directional-coupler network requiring only 2 couplers per lane. Finally, transmitting keyed hops $U_{0,1}[k] = (a_k, \pm b_k)/\sqrt{2} \in V_2(\mathbb{R}^7)$ creates our Zero-Divisor Keyed Modulation (ZDKM) LPI/LPD waveform (99.90% keyed Rx vs. 49.76%

radiometer) alongside our co-designed $8 \oplus 4 \oplus 4$ multiplier-free transceiver ($J^2 = -4\mathbf{I}_4$).

Table 1: Verified Baseline Performance Across Native 16D Sedenion Hardware Primitives

NSSP Hardware Primitive	Classical Baseline	Native Sedenion (Ours)	Verification Status
Pre-ADC Crossbar (+80 dB JSR)	−46.3 dB SINR (98.7% Clipped)	+21.3 dB SINR (68.4–82.8 dB Null)	84-Blade SPICE Sim
Co-Linear Jammer ($\Delta\theta = 0.0°$, +80 dB)	−80.0 dB SINR / 126,997 m CRLB	0.0000 dB Sat Loss / 17.96 m CRLB	Lean 4 + Passive S_{32}
Coherent Spoofer Det. (Rank = 1)	9.38% P_{det} ($K = 64$ Cov)	99.80% P_{det} ($K = 1$ Snap)	Lean 4 + Monte Carlo
Xilinx RFSoC `XCZU28DR` `Q1.14` Core	18,400 LUTs / 64 DSP48 / 140 cyc	512 LUTs / 0 DSP48 / 1 cyc (75.1 dB)	SystemVerilog RTL
Bare-Metal MCU & 4D Hinge Conditioner	1.46 / 71.2 μs Jitter; DMA Bug	0.27 / 6.9 μs Jitter (5.4–10.3×); 84.9 dB	Live `STM32WB55` + `ESP32-S2`

Live Multi-Band Bare-Metal Hardware Reduction to Practice: Beyond SPICE and RTL, we deployed a zero-heap (0 B `malloc`, 1.4 KB Flash ROM) bare-metal C/Rust NSSP engine on physical `STM32WB55RG` (Cortex-M4F @ 64 MHz), `ESP32-S2-WROVER` (Xtensa LX7 @ 240 MHz), and ARM64 NEON (2.44 μs/Moreno call, 84.2–87.0 dB `Q1.14` null depth). Tested across 2,797 live Sub-GHz `CC1101` captures (71,454 pulse pairs), 1,258 Optical IR bursts (83,377 pulses), 2,738 802.11 Wi-Fi trajectories, and 125 kHz/13.56 MHz inductive coils, our Lean 4-certified 2-Stage Glitch Coalescer and 4D Quaternion Hinge Projector (`MaxwellExterior.lean` §5.D) cuts Sub-GHz RMS timing jitter by $\mathbf{5.41}\times$ ($1.46 \to 0.27\,\mu$s), cuts IR jitter by $\mathbf{10.28}\times$ ($71.15 \to 6.92\,\mu$s), eliminates `0xFFFFFFFF` DMA timer reload underflow on $< 8\,\mu$s inductive glitches, and discriminates cross-band 13.56 MHz reader leakage ($\kappa_{\mathbb{S}} \in [0.098, 0.523]$) in $K = 1$ snapshot.

3. 12-Month Hardware Build Plan, Deliverables, and Transition

Why Proposed Work Is Not Yet Completed: While the Lean 4 theorems, 84-blade SPICE netlist (`crossbar_84_blades_rom.json`), and `Q1.14` SystemVerilog RTL (`sedenion_crpa_beamformer.sv`, 75.14 dB digital null depth) are complete, fabricating and testing the physical RF/analog boards requires three 12-month R&D tasks:

1. Task 1 (Mos. 1–4): Dyadic 2×2 Dual-Pol TEM Array & Passive Octonion Hybrid Feed PCB. Fabricate the 4-layer Rogers `RO4350B` antenna/balun feed board (`CGGBP.35.6.A.02` patches + `QCN-19+`/`TC1-1-13M+` hybrids) and characterize anechoic $V_2(\mathbb{R}^7)$ orthogonality. Deliverable D1: Calibrated Dyadic TEM Vector Array hardware & S-parameter report.

2. Task 2 (Mos. 5–8): Pre-ADC Analog Sedenion Crossbar Mezzanine Fabrication. Fabricate and populate the 16-channel differential analog crossbar PCB (`ADG1611` switches + `AD8130` summing amps + ±0.05% `ERA-3AEB1001V` networks) controlled by the Zynq FPGA $K = 1$ Associator Spoofer Detector. Deliverable D2: Bench-tested Pre-ADC Crossbar Mezzanine demonstrating > 65 dB pre-ADC analog jammer suppression prior to 12-bit ADC quantization.

3. Task 3 (Mos. 9–12): Over-the-Air ZDKM LPI/LPD & +80 dB JSR HIL Demonstration. Execute full hardware-in-the-loop (HIL) testing of +80 dB JSR pre-ADC nulling, $K = 1$ zero-lag repeater spoofer rejection, and ZDKM LPI/LPD transceiver links. Deliverable D3: Final Technical Report, Gerber/BOM package, and Government Purpose Rights (GPR) hardware/RTL deliverables for U.S. DoD transition.

Price Realism / Rough Order of Magnitude (ROM)

(Per DARPA-PS-26-09 Section 1.2, the Price/ROM section is $\leq \frac{1}{2}$ page and does not count against the 3-page written abstract limit.)

The total Rough Order of Magnitude (ROM) for the 12-month hardware prototype and RF validation effort is $485,000.00 USD. All foundational Lean 4 proofs, 84-blade crossbar netlists, and baseline `Q1.14` SystemVerilog RTL cores are leveraged at zero cost to the U.S. Government.

Resource Category / Labor & Materials	Basis of Estimate (12 Mos)	Cost (USD)
Principal Investigator / Sedenion RF-DSP Architect (Pre-ADC crossbar architecture, $K = 1$ associator spoofer detector, ZDKM LPI/LPD transceiver integration)	1,200 Direct Labor Hours @ $150/hr (loaded)	$180,000.00
Senior RF / Mixed-Signal PCB Design Engineer (Rogers RO4350B dyadic TEM hybrid feed, 16-ch differential `ADG1611`/`AD8130` analog crossbar layout & SI)	1,000 Direct Labor Hours @ $140/hr (loaded)	$140,000.00
FPGA & HIL Test Engineer (Zynq GPIO crossbar controller, 16-ch phase-coherent SDR interface, anechoic & +80 dB JSR HIL testing)	600 Direct Labor Hours @ $115/hr (loaded)	$69,000.00
PCB Fabrication, Precision Components & RF Testbed (4-layer Rogers/FR4 PCBs, ±0.05%/±0.01% resistor arrays, 16-ch phase-coherent SDR bank, VNA/chamber rental)	Vendor BOM & Fab Quotes	$76,000.00
Travel, Reporting & DoD Transition Reviews (2× technical review trips to DARPA STO, Arlington, VA)	2 Trips + Hardware Demo	$20,000.00
Total Proposed Rough Order of Magnitude (ROM)	2,800 Hrs + Hardware	$485,000.00